CP610 Project Deliverable #4 (Worth 6/30)

Due 30 Nov, 2025 at 11:59pm

<u>What to Do:</u>

1. Using the dataset from Deliverable #2, build the following machine learning models:

   Regression

   - Use the 2023 monthly sales data to train a Polynomial Regression model
   - Test the model on the 2024 and 2025 monthly sales
   - Experiment with polynomials of varying degrees and pick the best degree
   - Experiment with various learning rates and pick the best learning rate
   - Make sure to avoid overfitting

   Clustering

   - Cluster 20% of randomly selected customers based on Number of Transactions and Total Spent using K-Means
   - Experiment with different Ks and pick the K you believe is the best
   - Explain what the clusters represent (i.e., give them names)

2. Build and test these models using Python libraries, and using Excel

<u>What to deliver:</u>

Place the following in a zip file. The file name should follow the format deliverable#4-your-group-name. Submit the zip file at the Deliverable #4 Dropbox on MyLS.

1. An excel workbook that contains all your work.
2. The commented Python code you used to perform the task.
3. A report that explains what you did in both Excel and Python. and the rationale behind the decisions you made throughout the process.
4. <u>Make your report self-sufficient. I should be able to learn about what you did without having to use the other documents you submitted.</u>

Note:

No email submissions will be accepted. Start early and submit your deliverable on time!